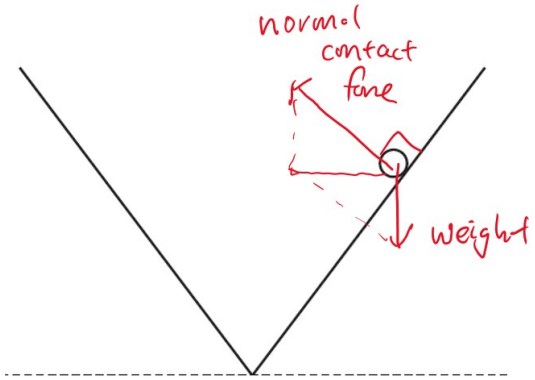
H2 Physics Prelim Exams 2026 – Paper 2 Solutions

1(a)	rate of change of velocity	[1]
1(b)	Gain in KE = Loss in GPE $\frac{1}{2}m(v_f^2 - v_i^2) = mg(h_i - h_f)$ $v^2 = 5.9^2 + 2 \times 9.81 \times 7.8$ $v^2 = 187.846$ $v = 13.711 = 13.7$ or 14 m s^{-1} or by resolving components Vertically: $\uparrow: v^2 = u^2 + 2as$ $v^2 = (5.9 \sin 60^o)^2 + 2(-9.81)(1.2-9.0)$ $v_v = 13.3844 = 13.4$ Horizontally: $v_h = 5.9 \cos 60^o$ $v_h = 2.95$ Resultant velocity, $v = \sqrt{(13.4^2 + 2.95^2)} = 13.72 = 13.7$ or 14 m s^{-1}	[1] [1] [1] [1] [1] [1]
1(ci)	$W = mg = 78 \times 9.81 = 765.18 = 765$ N Resultant Force = 740 + 950 - 765 = 925 N Acceleration = F/m = 925/78 = 11.9 m s^{-2}	[1] [1] [1]
1(cii)	(As the diver moves down their) speed decreases (So) viscous force / drag (force) decreases	[1] [1]

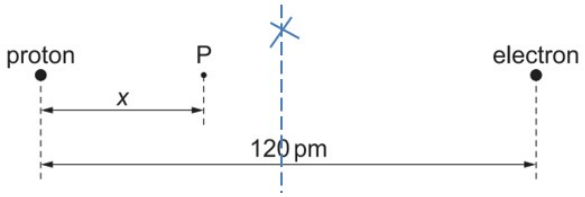
2(a)	 correct arrow direction – vertically downwards arrow for weight and arrow perpendicular to the cone**,** inwards and upwards for normal contact correct naming for both forces – weight and normal contact force/ normal force	[1] [1]
2(b)	vertical component of the normal contact force = weight, thus F_{net} = 0 in the vertical direction horizontal component of the normal contact force is resultant force towards centre (of circle)	[1] [1]
2(c)	N cos52^o = W = mg ----- (1) N sin52^o = mv^2/r ----- (2) (2)/(1): tan 52° = v^2/gr	[1] [1]

	$v = gr \tan 52^o = 9.81\ (0.15)(\tan 52^o)$ $= 1.372$ $= 1.4\ m\ s^{-1}$ (shown)	[1]
2(d)	$v = r\omega$ or $a = r\omega^2$ $\omega = 1.4 / 0.15$ or $\sqrt{(9.81 \times \tan 52° / 0.15)}$ $= 9.3\ rad\ s^{-1}$ or $9.149 = 9.1\ rad\ s^{-1}$	[1]
2(e)	Since $\tan 52^o = v^2/gr$, v^2 is proportional to r so if speed increase, radius must also increase or (same mass) resultant/centripetal force/acceleration ($N \sin 52^o$ and $N = W/\cos 52^o$) = constant, since $F = mv^2/r$ or $a = v^2/r$, v^2 is proportional to r so if speed increase, radius must also increase	[1] [1] [1] [1]

3(ai)	Gas is ideal as the graph is a straight line through the origin, satisfying the ideal gas equation $pV = NkT$	[1]
3(aii)	$pV = NkT$ $N = pV / kT$ = gradient of graph $= 270 / 8.0 \times 10^{-21}$ or substitute a data point	[1]

	$n = N/N_A$ $= (3.4 \times 10^{22})/ (6.02 \times 10^{23})$ $= 0.056$ mol	[1]
3(b)	½ m$<c^2>$ = 3/2 kT When pV = 270 J, $kT = 8.0 \times 10^{-21}$ J ½ m $(1900^2) = 1.5 (8.0 \times 10^{-21})$ m = 6.65×10^{-27} kg $= (6.65 \times 10^{-27}) / (1.66 \times 10^{-27})$ $= 4.0\ u$	[1] [1] [1]

4(a)	work done per unit charge by an external force In bringing a small positive test charge from infinity to that point	[1] [1]
4(bi1)	potential (due to proton) = $\dfrac{1.6 \times 10^{-19}}{4\pi(8.85 \times 10^{-12})(10 \times 10^{-12})}$ = 143.86888 potential (due to electron) = $\dfrac{-1.6 \times 10^{-19}}{4\pi(8.85 \times 10^{-12})(110 \times 10^{-12})}$ = 13.07899 V = 143.86888 - 13.07899 = 130.7 ≈ 130 V (2sf)	[1] [1]

4(bi2)	$V_P = \frac{Q_{proton}}{4\pi\varepsilon_o x} + \frac{Q_{electron}}{4\pi\varepsilon_o(120 \times 10^{-12} - x)}$ $= \frac{1.6\times10^{-19}}{4\pi(8.85\times10^{-12})(30\times10^{-12})} + \frac{-1.6\times10^{-19}}{4\pi(8.85\times10^{-12})(90\times10^{-12})}$ = + 31.91 V = 32 V	[1]
4(bi3)	cross drawn midway, equidistant between the electron and the proton  	[1]
4(bi4)	 starting from (10, +130) to (110, –130) curve decreasing gradient until x = 60 pm, crossing V = 0 at (60, 0) and then getting steeper after x = 60 pm	

	curve passing through (30, +32) and (90, -32)	[1] [1] [1]
4(bii)	$E_{midway} = \frac{Q_{proton}}{4\pi\varepsilon_o x^2} + \frac{Q_{electron}}{4\pi\varepsilon_o x^2}$ $= \frac{1.6\times10^{-19}}{4\pi\left(8.85\times10^{-12}\right)\left(60\times10^{-12}\right)^2} + \frac{1.6\times10^{-19}}{4\pi\left(8.85\times10^{-12}\right)\left(60\times10^{-12}\right)^2}$ = 8.0 x 10^{11} N C^{-1} (to the right)	[1] [1]
4(biii)	Rotate clockwise Accelerate upwards	[1] [1]

5 (a)	The magnetic flux density is defined as the force acting per unit current per unit length on a conductor placed perpendicular to the magnetic field	 [1] [1]
(b)(i)	Protons are positively charged thus (current is in direction of the motion of the proton) and magnetic force towards the centre of the circular path (may describe direction of magnetic force at a specific point) By Fleming left hand rule, using the direction of current and force, the magnetic field must be out of the paper.	[1] [1]
(ii)	The centripetal force is provided by the magnetic force $F_B = F_C$ $Bqv = mv^2/r$ $v = Ber/m$ $\frac{1}{2} mv^2 = \frac{1}{2} m (Ber/m)^2$ $KE = \frac{B^2 e^2 r^2}{2m}$	[1] [1]
(iii)	E-field direction is in same direction to the motion of the proton whenever the proton crosses the gap. The electric force acting on the proton causes it to accelerate continuously each time it crosses the gap thus KE increases.	[1] [1]

6(a)(i)	From Faraday's equation, Emf = - change in flux linkage / time taken, $\Rightarrow$ d*t* = - change in flux linkage / emf $= - (+8.0 \times 10^{-5}) / (-5.0 \times 10^{-4})$ $= 0.16$ s Speed = displacement / *dt* $= 0.03 / 0.16 = 0.1875 = 0.19$ m s^{-1}	[1] [1]
(ii)	Increase in flux linkage = $N \times (5 - 3) \times 10^{-6} = 8.0 \times 10^{-5}$ Wb-turns $\Rightarrow N = 40$ turns	[1]
(iii)	Since the coil moves at constant speed, Ɛ = constant and Ɛ = - dΦ / dt thus - dΦ / dt = constant, the Φ vs t graph should have constant gradient. From P to x = 3.0 cm, At P, magnetic flux = 3×10^{-6} Wb (given in part (ii)), thus magnetic flux linkage = N(magnetic flux) = 40 $(3 \times 10^{-6}) = 1.2 \times 10^{-4}$ Wb	

turns

The magnetic flux linkage increases by 8.0 x 10^{-5} Wb-turns from P to x = 3.0 cm (given in part (i)).

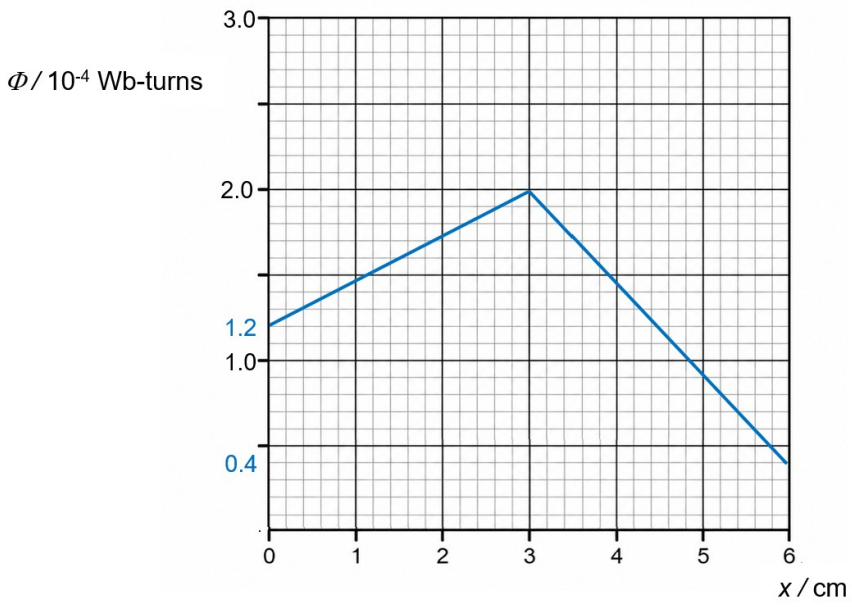
Thus final magnetic flux linkage at x = 3.0 cm is 1.2 x 10^{-4} + 0.8 x 10^{-4}

= 2.0 x 10^{-4} Wb-turns.

This is a positive gradient to show an increase in flux linkage.

From x = 3.0 cm to x = 6.0 cm,

Since Ɛ is doubled and changed in sign from -5 x 10^{-4} V to +10 x 10^{-4} V and Ɛ = - dΦ / dt , the gradient in this region should be doubled and negative to show a decrease in flux linkage.



This mark is for those who read ai) and noticed -5 x 10^{-4} V n +10 x 10^{-4} V in emf graph

- inverted V shape
- gradient of 2nd segment = twice of 1st segment [both hv equal x-axis so check the change in Φ is twice (16 small sqs) in segment 2 compared to segment 1 (8 small sqs)]

	● the change in Φ for 1st segment must be the value in ai) i.e. 8.0 x 10^{-5} Wb turns This mark is for students who used the values in aii) to calculate the initial Φ ● correct initial flux linkage at 1.2 x 10^{-4} Wb-turns	[1] [1]
(b)	By **Faraday's law**, ring **cuts the magnetic flux/field lines** and an **e.m.f. is induced** in the ring. Induced e.m.f. causes **induced/eddy currents to form** in the ring. Either, By **Lenz's law,** the **induced currents flow** in such a direction so as to oppose the change in flux/ flux cutting the ring, So an resistive/opposing force is induced/produced on the ring to oppose its	[1] [1] [1]

	motion. **Or** Currents in the ring **dissipates thermal energy,** And this thermal energy **comes from/is converted from the mechanical energy** of the oscillations.	[1] [1] [1]

7(a)	tungsten has a very high melting point	[1]
7(b)	Using $I / I_0 = e^{(-\mu x)}$ $0.5 = e^{(-2.46 \times x)}$ $x = \ln(2) / 2.46 = 0.282$ cm	[1] [1]
(c)(i)1	From Table 7.1, μ of muscle = 0.40 cm^{-1} Using $I / I_0 = e^{(-\mu x)}$ ratio along MM = $e^{(-0.40 \times 8.0)}$ = 0.041 or 0.0408	[1]
(c)(i)2	From Table 7.1, μ of bone = 2.46 cm^{-1} Using $I / I_0 = e^{(-\mu x)}$ For 2 cm of muscle, $I / I_0 = e^{(-0.40 \times 2.0)}$ or For 4 cm of muscle, $I / I_0 = e^{(-0.40 \times 4.0)}$ For 4 cm of bone, $I / I_0 = e^{(-2.46 \times 4.0)}$ Ratio along BB = (I / I_0)_muscle_2cm × (I / I_0)_bone × (I / I_0)_muscle_2cm = $e^{(-0.40 \times 2.0)} \times e^{(-2.46 \times 4.0)} \times e^{(-0.40 \times 2.0)}$ = 0.0000108 or 0.000011 or	[1 - for each I / I_0] [1]

	Ratio = e^(-0.40 × 4.0) × e^(-2.46 × 4.0) = 0.0000108 or 0.000011	
(c)(ii)	Since the ratio along MM is 0.041/0.000011 = 3727/ ≈ 10^3/ is about 3 orders of magnitude bigger/different than that along BB, the contrast is good.	[1] [1]
7(d)	To produce additional magnetic fields that cause the B to vary with position across patient This can be achieved when the magnetic field in the gradient coils superposed with the magnetic field from the main coil to produce varying magnetic fields at different positions.	[1] [1]
(e)	MRI is able to provide excellent contrast between the different parts of the brain (gray matter, white matter and blood) due to the distinct relaxation times in Table. 7.2, whereas X-ray has poor contrast between blood and brain due to similar absorption coefficients in Table 7.1	[1] [1]
(f)	(Any two relevant point) ● small space, not suitable for patient who are claustrophobic ● metal implants (e.g. pacemaker, cochlear implants) / ferromagnetic implants in patients may malfunction / move in the strong magnetic field ● (long scan time) patient needs to remain still for long duration ● any movement may cause blurred image, children or patients who cannot lie down for long duration and any reasonable challenges like	[2]

	loud noise which may be scary for some patients ● high cost	
(g)	For a solenoid, $B = \mu_0 n I$ $1.5 = (4\pi \times 10^{-7}) \times n \times 750$ $n = 1591.55\ m^{-1}$ number of turns = n × 2.2 = 3500 turns	[1] [1]
(h)	Larmor frequency, $\omega = \gamma B_0$ $= (2.68 \times 10^8) \times 1.5 = 4.02 \times 10^8$ rad s^{-1} frequency of RF wave = $\omega / 2\pi$ $= 4.02 \times 10^8 / 2\pi$ $= 6.4 \times 10^7$ Hz	[1] [1]
(i)	Assuming all electrical energy is used to vaporise liquid helium: rate of electrical heating = rate of vaporisation of liquid helium $I^2 R = m_rate \times L_v$ (where m_rate is the initial rate of vaporization) $(750)^2(0.0045) = m_rate\ (21 \times 10^3)$ m_rate = 0.12 or 0.121 kg s^{-1}	[1] [1]